

INTERIM PROGRESS REPORT

on

AI-based Urban Traffic Management System using Reinforcement Learning and Computer Vision

Submitted by

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Under the supervision of

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Abstract

Urban traffic congestion is a major challenge in modern cities, leading to increased travel time, fuel consumption, and safety risks. Traditional traffic signal systems operate on fixed timings and fail to adapt to real-time conditions such as varying traffic density, pedestrian movement, weather changes, and emergency vehicle requirements.

This project proposes an AI-based Urban Traffic Management System using Reinforcement Learning (RL) and Computer Vision (CV) to enable adaptive and intelligent traffic signal control. Computer Vision is used to detect vehicles, pedestrians, and emergency vehicles from real-time camera feeds, while Reinforcement Learning dynamically adjusts signal timings based on traffic conditions.

The system aims to improve traffic flow, reduce waiting time, prioritize emergency vehicles, and enhance pedestrian safety. It also helps in reducing fuel consumption by minimizing unnecessary idling at signals.

This project supports the United Nations Sustainable Development Goals (SDGs), particularly:

SDG 9: Industry, Innovation and Infrastructure – The project uses advanced technologies like Artificial Intelligence to improve existing traffic systems. It promotes innovation and supports the development of smarter and more efficient infrastructure.

SDG 11: Sustainable Cities and Communities – Modern cities face serious traffic problems that affect daily life. By reducing congestion and improving traffic flow, this system helps make cities more organized, safer, and easier to move around in.

SDG 13: Climate Action – Traffic congestion leads to unnecessary fuel consumption and increased pollution. By reducing waiting time at signals and improving traffic movement, the system helping to lower emissions and hence it also contributes to a cleaner environment.



Literature Survey

TITLE & JOURNAL	AUTHOR	YEAR	CONTENT
Intelligent Traffic Signal Control Based on Reinforcement Learning with Edge Computing and Intelligent Reflecting Surface <i>Zhang, G., Yue, Y., Yang, J., & Tang, J. (2024) — IEEE Access</i>	Zhang, G., Yue, Y., Yang, J., & Tang, J.	2024	Combines reinforcement learning and edge computing for faster real-time traffic decisions. Supports intelligent and scalable traffic signal optimization.
Reinforcement Learning for Traffic Signal Control in Hybrid Action Space <i>Luo, H., & Jin, S. (2022) — IEEE Transactions on Intelligent Transportation Systems</i>	Luo, H., & Jin, S.	2022	Introduces an RL-based model that adapts signal timing in complex traffic scenarios. Helps reduce congestion and improve traffic management.
Learning Traffic Signal Control from Demonstrations <i>Zheng et al. (2020) — Proceedings of the 26th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)</i>	Zheng et al.	2020	Proposes learning traffic signal policies from expert demonstrations. Achieves stable and efficient signal control with reduced manual intervention.
Deep Learning for Real-Time Traffic Signal Control <i>Chen Chen (2017) — IEEE Intelligent Transportation Systems Magazine</i>	Chen Chen	2017	Applies deep learning for dynamic traffic signal control using real-time data. Enhances efficiency and adaptability in urban traffic systems.
Traffic Signal Timing via Deep Reinforcement Learning <i>L. Li (2016) — IEEE Transactions on Intelligent Transportation Systems</i>	L. Li	2016	Uses deep reinforcement learning to optimize traffic signal timing based on traffic conditions. Improves traffic flow and reduces waiting time compared to fixed signals.

System Model Description

The proposed system is an AI-based traffic management model that integrates Computer Vision and Reinforcement Learning for adaptive signal control.

The system consists of three main components: input layer, processing layer, and decision layer. In the input layer, real-time data is collected through traffic cameras and external sources such as congestion and weather data.

In the processing layer, Computer Vision techniques (using OpenCV and YOLO) are used to detect vehicles, pedestrians, and emergency vehicles from live video feeds. The system analyzes traffic density and movement patterns based on this data.

In the decision layer, a Reinforcement Learning (RL) model processes the analyzed data and determines the optimal traffic signal timing. It gives priority to emergency vehicles and ensures pedestrian safety while minimizing traffic congestion.

Finally, the output layer controls traffic signals dynamically based on the RL decisions, resulting in improved traffic flow, reduced waiting time, and lower fuel consumption.

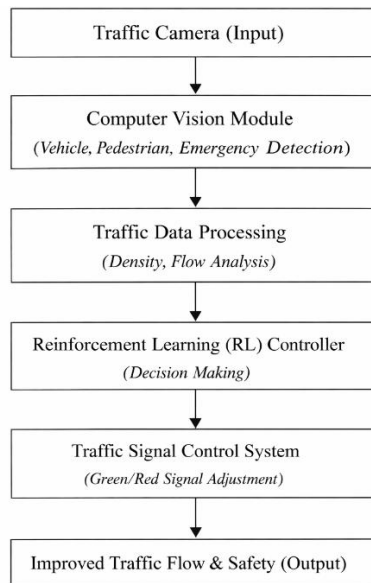


Fig 1 : Block diagram of the system model

Software Used

The system uses a combination of modern software tools to enable intelligent traffic control. Frameworks like Flask and SQLite are used for building a lightweight backend and storing traffic data efficiently. Computer Vision is implemented using OpenCV along with YOLOv8 for real-time detection of vehicles, pedestrians, and emergency vehicles from camera feeds. For decision-making, a Deep Q-Network (DQN) based Reinforcement Learning model is developed using TensorFlow, which dynamically adjusts traffic signal timings based on traffic conditions. Additionally, Raspberry Pi GPIO libraries are used to interface software logic with physical traffic signals.

Components (Software):

- Flask
- SQLite
- YOLOv8
- OpenCV
- TensorFlow (DQN)
- Raspberry Pi (RPi.GPIO)

Hardware Used

The hardware setup is designed to simulate and implement a real-time traffic intersection. A Raspberry Pi acts as the central processing unit, handling both image processing and control logic. USB cameras capture live traffic footage, while IR sensors are used to detect vehicle presence in each lane. Traffic light modules represent the signal system, controlled based on decisions from the AI model. Supporting components such as jumper wires and a breadboard are used for circuit connections, and mini cars are included to demonstrate real-world traffic scenarios in a prototype environment.

Components (Hardware):

- Raspberry Pi
- USB Camera
- IR Sensors (×4)
- Traffic Light Modules (×4)
- Jumper Wires
- Breadboard

Methodology

- **Data Collection**
Traffic data is captured using cameras at road intersections.
- **Image Processing (Computer Vision)**
OpenCV and YOLO are used to detect vehicles, pedestrians, and emergency vehicles.
- **Traffic Analysis**
The system calculates traffic density and movement patterns from the detected data.
- **Decision Making (Reinforcement Learning)**
RL model analyzes traffic conditions and decides optimal signal timing.
- **Signal Control**
Traffic lights are adjusted dynamically based on RL decisions.
- **Continuous Learning**
The system learns from new traffic data and improves performance over time.

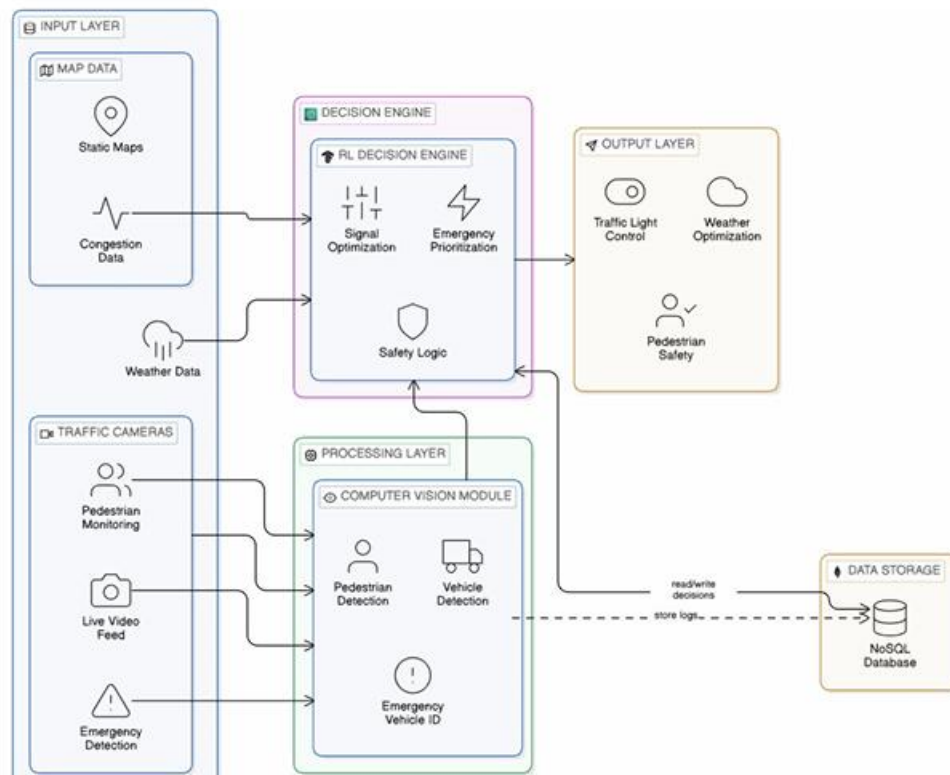


Fig 2 : Methodology / Workflow

Statement of Contribution

This interim progress report, and the activities that it has entailed, has been a collaborative effort amongst the group members.

Amit Chauhan (RA2311004010332)

- Designed system architecture and overall workflow of the project
- Worked on integration of Reinforcement Learning model for signal control
- Assisted in hardware and software tool selection (Raspberry Pi, OpenCV, YOLO)

Shaily Giri (RA2311004010347)

- Prepared the project report and documentation
- Designed and developed the presentation (PPT)
- Worked on concept development and system flow (CV + RL interaction)

Lakshita (RA2311004010354)

- Conducted literature survey and research analysis
- Assisted in implementation planning for Computer Vision (vehicle & pedestrian detection)
- Supported data handling and project coordination

Declaration

We hereby declare that the project titled “*AI-based Urban Traffic Management System using Reinforcement Learning and Computer Vision*” is a Bonafide work carried out by us under the guidance of **Dr. Bharatha Babu K**

This work is submitted as part of our academic requirements and has not been submitted elsewhere for any other degree or certification. All sources of information used in this report have been properly acknowledged.

Project Evaluator: Mr. M. Aravindan

Amit Chauhan (RA2311004010332) :

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Project Supervisor:

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INTERIM PROGRESS REPORT EVALUATION RUBRICS

Project Title : AI-based Urban Traffic Management System using Reinforcement
Learning and Computer Vision

Project Code : 21ECP302L

Team Members: Amit Chauhan (RA2311004010332)
Shaily Giri (RA2311004010347)
Lakshita (RA2311004010354)

Name of Evaluator: Mr. M. Aravindan

Rubrics	Possible Points	Assigned Marks		
		Reg No. (332)	Reg No. (347)	Reg No. (354)
Aim & Objectives, Abstract Objective complete and well-written provides all necessary background principles for the experiment	5			
Content-Introduction, Literature review, System model Technically correct; Contain in-depth and complete details of the project	10			
Language (Word Choice, Grammar) Sentences are complete and grammatical. They flow together easily Words are chosen for their precise meaning. Engineering terms and jargon are used correctly; No misspelled words. Numerical Usage and Illustrations All figures, graphs, charts, and drawings are accurate, consistent with the text, and of good quality. They enhance understanding of the text. All items are labelled and referred to in the text. All equations are clear, accurate, and labelled. All variables are defined and units specified. Discussion about the equation development and use is stated.	5			
Total Points	20			

Date:

Signature of the Project Guide